

ENTERPRISE SECURITY ASSESSMENT LAB

AD NETWORK SECURITY METHODOLOGY REPORT

Phase 5 - Requesting service tickets for SPN-enabled accounts

Report Section	Phase 05 - Kerberoasting SPN Ticket Request
Project Area	Internal Network and Active Directory Security Assessment
Environment	Private Windows AD lab - Kali Linux, Windows Server, Windows endpoint
Prepared For	GitHub Portfolio Documentation
Document Type	Individual methodology PDF with redacted evidence

Authorized lab documentation only. Sensitive credential material, hashes, passwords, and replication output have been redacted before publication.

1. Executive Summary

This report documents **Kerberoasting SPN Ticket Request** as part of the AD Network module in the Enterprise Security Assessment Lab. The section is written as a professional, portfolio-safe methodology report that explains the testing objective, workflow, evidence, tool usage, sanitized output, security risk, remediation guidance, and GitHub publication requirements.

Objective: Identify SPN-enabled accounts and request Kerberos service ticket material that can be tested offline for weak service account passwords.

MITRE / Framework Mapping	T1558.003 - Steal or Forge Kerberos Tickets: Kerberoasting
Testing Scope	Authorized private Active Directory lab only
Evidence Handling	Screenshots embedded with necessary redaction; raw hashes and secrets are not included

2. Methodology and Workflow

The methodology below is designed to show a controlled penetration-testing workflow rather than random tool execution. Each step supports a specific assessment goal and produces evidence that can be reviewed safely in a public portfolio.

- Use a valid low-privileged domain credential inside the lab.
- Run GetUserSPNs to identify accounts with Service Principal Names.
- Request TGS material only for the authorized lab domain.
- Save ticket material locally and do not upload raw hash files.
- Use the output as input for offline Kerberoast cracking validation.

Command / Workflow Used

```
# Identify and request Kerberoastable tickets using Impacket
GetUserSPNs.py <DOMAIN>/<USER>:<PASSWORD> -dc-ip <DC_IP> -request -outputfile kerberoast_hashes.txt
```

Sanitized Output Style

```
ServicePrincipalName: MSSQLSvc/<HOST>:1433
Account: <SERVICE_ACCOUNT>
Ticket material: <REDACTED_KERBEROAST_HASH>
Output file: kerberoast_hashes.txt (not uploaded to GitHub)
```

3. Evidence Mapping and Screenshot Explanation

- Avoid privileged service accounts unless absolutely required.
- Monitor abnormal Kerberos TGS request patterns.
- Review SPNs and remove unnecessary service registrations.

7. GitHub Redaction and Publication Guidance

- Publish only redacted evidence. Do not upload raw hashes, password material, ticket files, DCSync output, or secretsdump text dumps.
- Keep this report under 03-ad-network/reports/methodology/ and keep raw tool output in a private working folder unless fully sanitized.
- Use placeholders such as , , , and in public documentation.
- Confirm the screenshot file name in the README matches the repository file name exactly, including case, hyphens, and extension.

Do Not Upload

```
*.hash
*_hashes.txt
responder logs with raw NetNTLMv2
kerberoast_hashes.txt
secretsdump output
DCSync dumps
cracked-password files
unredacted BloodHound collection ZIPs if they expose sensitive names
raw command history containing passwords
```

8. Conclusion

The **Kerberoasting SPN Ticket Request** phase demonstrates a structured and evidence-based AD testing workflow. The report explains the purpose of the test, shows sanitized evidence, connects the activity to real-world risk, and provides practical remediation guidance. This makes it suitable for a professional cybersecurity GitHub portfolio.